Translations of tangent functions

1 **B**

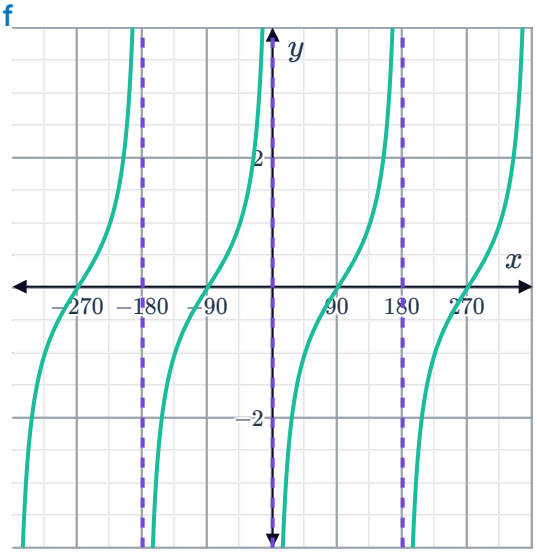
2 Vertical translation by 4 units upwards.

3 True

4

a No **b** 180° **c** 180° **d** 180°

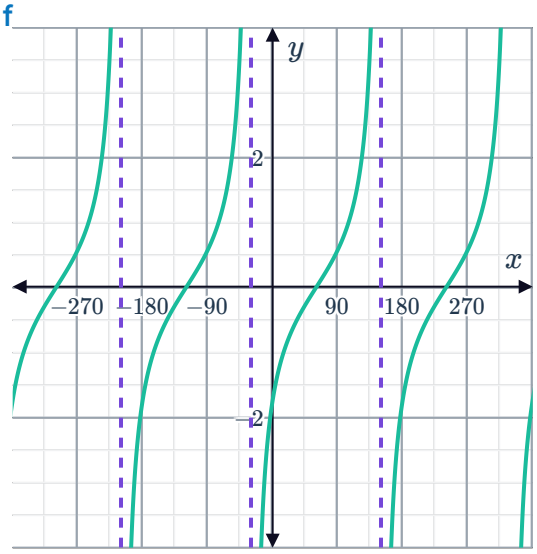
e 0 **f**



5

a $-\sqrt{3}$ **b** 180° **c** 180° **d** 150°

e $(-30)^\circ$ **f**



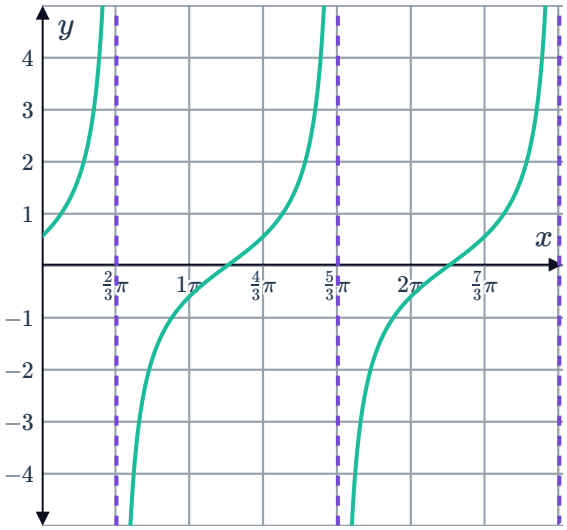
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6

a $x = \frac{2\pi}{3}, \frac{5\pi}{3}, \frac{8\pi}{3}$

b $x = \frac{7\pi}{6}, \frac{13\pi}{6}$

c



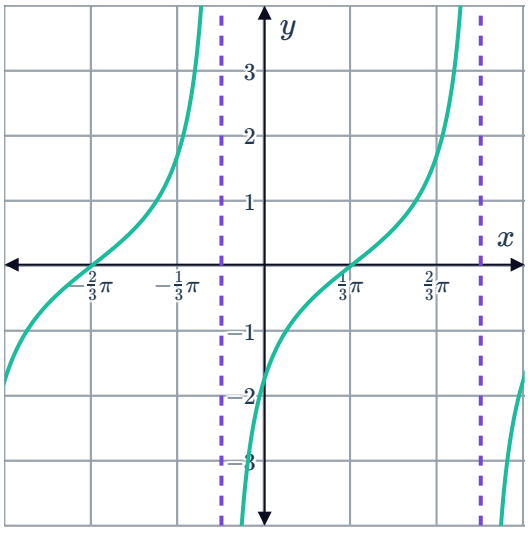
7

a

i $\left(\frac{\pi}{3}, 0\right)$ **ii** $\left(\frac{7\pi}{12}, 1\right)$ **iii** $\left(\frac{5\pi}{6}, 0\right)$

b Horizontal translation (phase shift) of $\frac{\pi}{3}$ units right

c



8

a

i Yes **ii** Yes **iii** Yes **iv** No

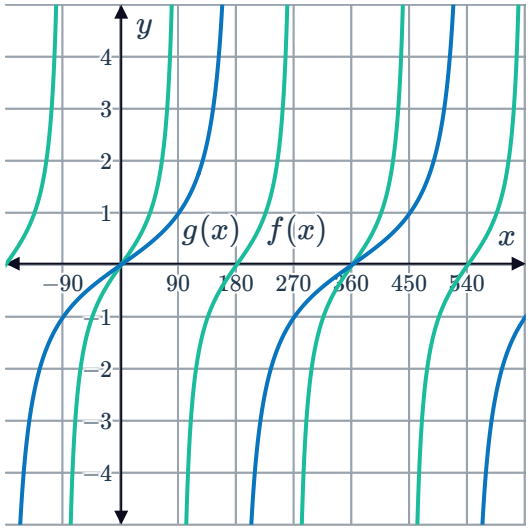
b **A**

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9 $f(x)$ and $g(x)$ have the same range, but different domains.
Subtracting 45° from x translates a function horizontally, which changes the position of the asymptote.

Dilations of tangent functions

10



11

a

Function	Period
$y = \tan x$	π
$y = \tan 2x$	$\frac{\pi}{2}$
$y = \tan 3x$	$\frac{\pi}{3}$
$y = \tan 4x$	$\frac{\pi}{4}$

b $\frac{\pi}{b}$

c The period becomes smaller.

12

a 60°

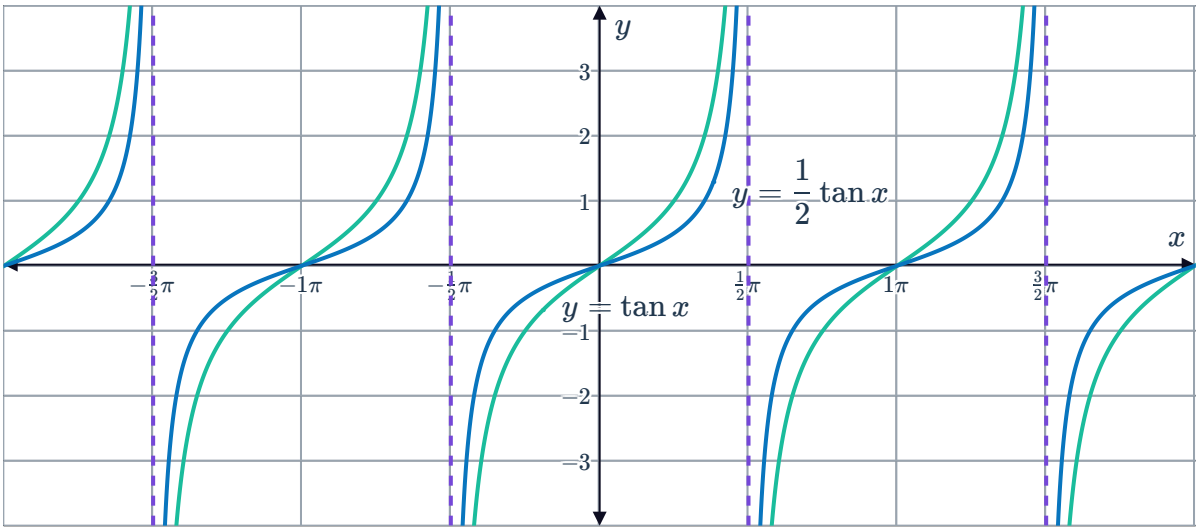
b $\tan 3x$

13 $g(x) = \tan(4x)$

14 Vertical stretch by a factor of 3.

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15



16 The graph of $y = \tan x$ has been stretched vertically by a factor of 5 and reflected across the x-axis.

17

a Reflecting the graph of $y = \tan x$ about the x-axis.

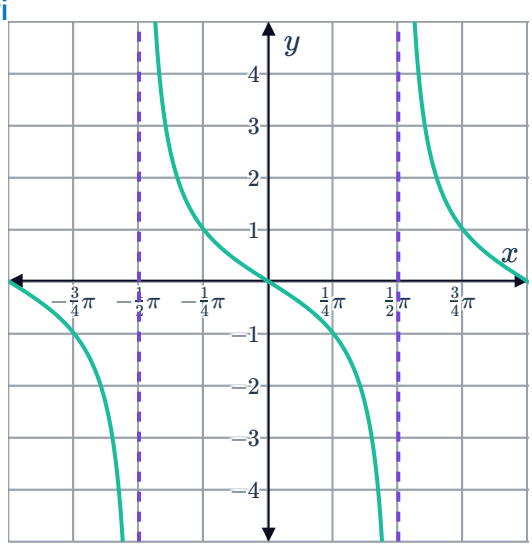
b $y = -\tan x$

18

a

i $y = 0$ **ii** π **iii** π **iv** $x = \frac{\pi}{2}$

v $x = -\frac{\pi}{2}$ **vi**

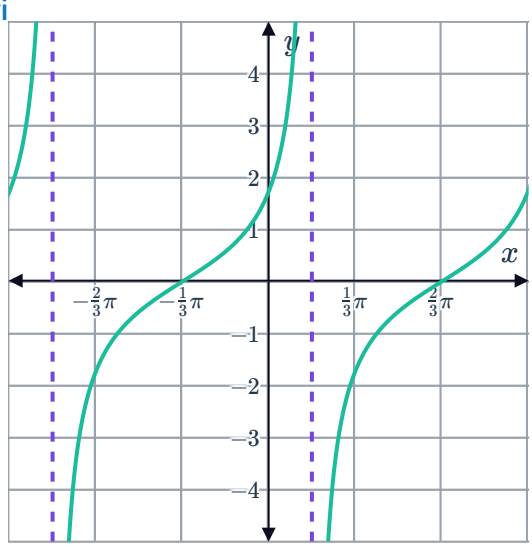


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b

i $y = \sqrt{3}$ **ii** π **iii** π **iv** $x = \dfrac{\pi}{6}$

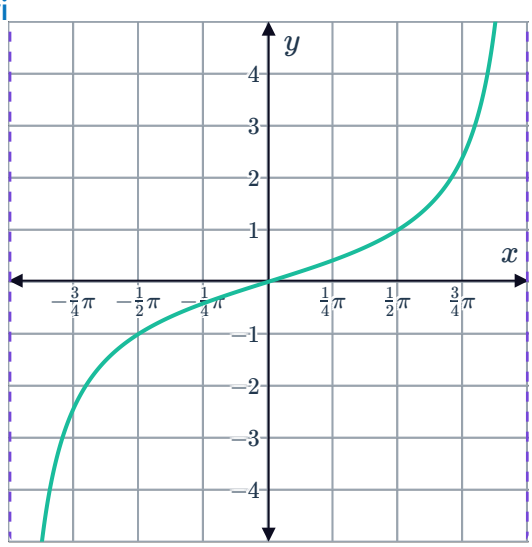
v $x = -\dfrac{5\pi}{6}$ **vi**



c

i $y = 0$ **ii** 2π **iii** 2π **iv** $x = \pi$

v $x = -\pi$ **vi**



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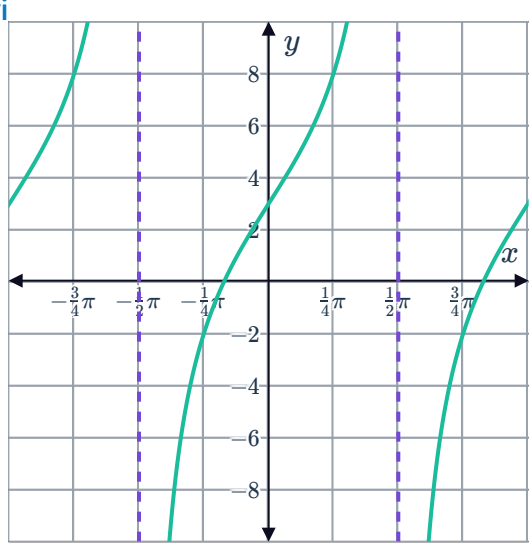
Series of transformations

19

a

i $y = 8$ ii π iii π iv $x = \dfrac{\pi}{2}$

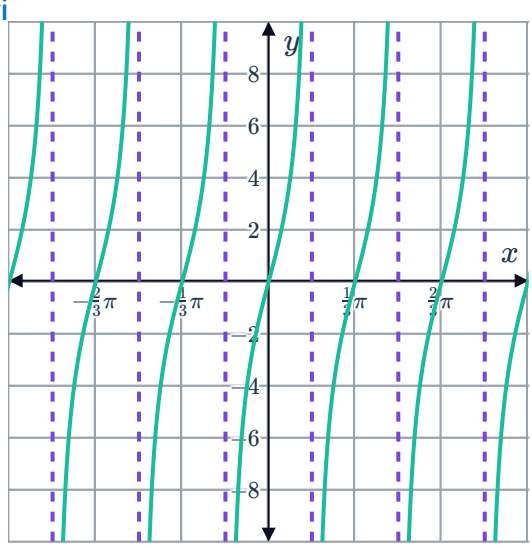
v $x = -\dfrac{\pi}{2}$ vi



b

i $y = -4$ ii $\dfrac{\pi}{3}$ iii $\dfrac{\pi}{3}$ iv $x = \dfrac{\pi}{6}$

v $x = -\dfrac{\pi}{6}$ vi

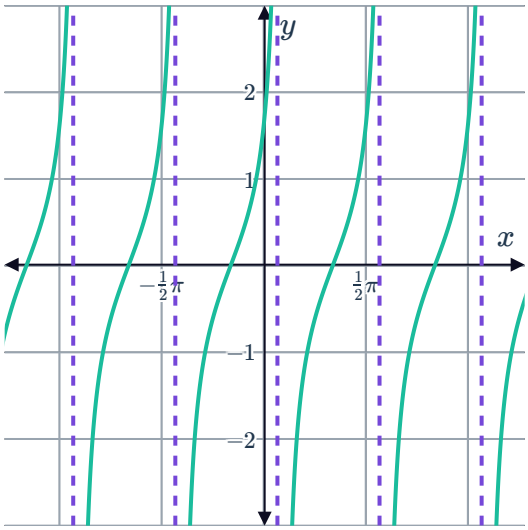


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20

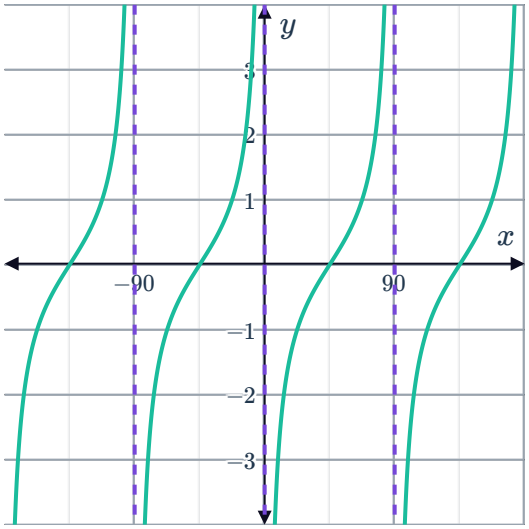
a $y = \tan\left(2\left(x + \frac{\pi}{6}\right)\right)$

b

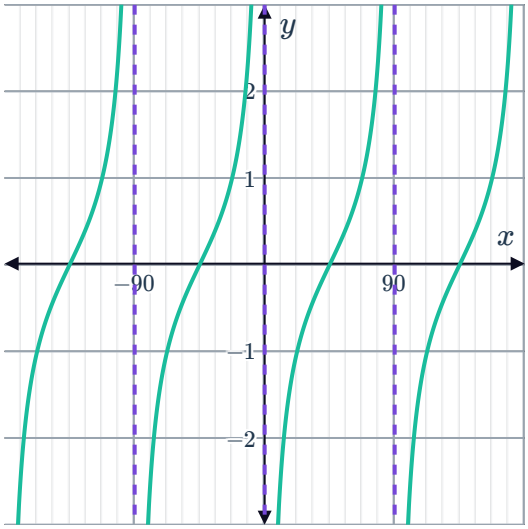


21

a



b



22

a No **b** Yes **c** No **d** Yes

23 $y = \tan(-x) - 3$

24

a Compressing the graph of $y = \tan x$ horizontally, and translating it vertically.

b $y = \tan 4x + 4$

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25

a All real x except when $x = \pi k + \frac{\pi}{2}$ for all integers k.

b $(-\infty, \infty)$

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